

Harry & Mae's Inc. Security Architecture

AARON MARTINEZ

SECURITY ASSESSMENT & RECOMMENDATIONS

Introduction

Aaron Martinez Solutions (AMS)

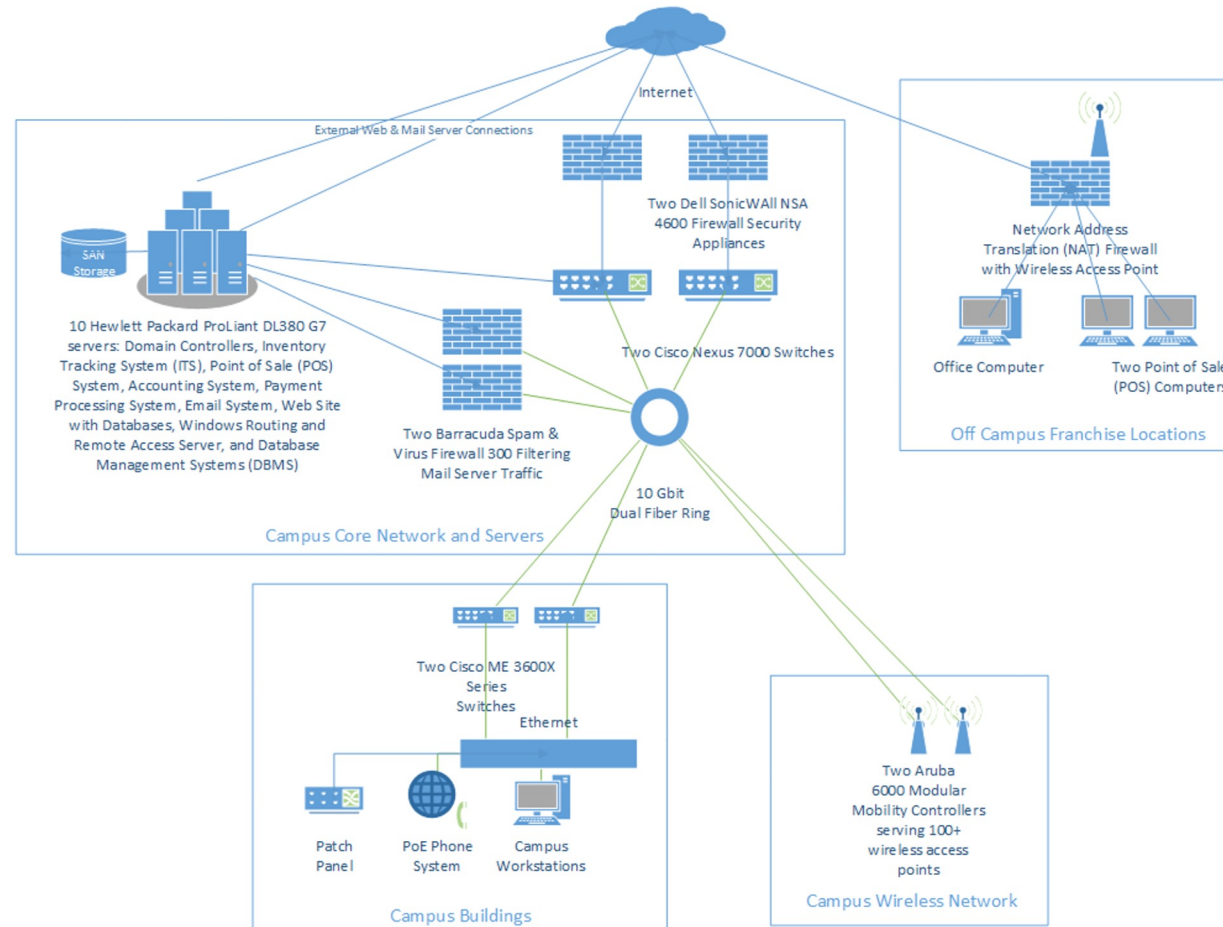
13+ Years Professional IT Experience:

- ▶ Software Development
 - ▶ Full Stack
 - ▶ Cross-Platform
- ▶ System Administration
 - ▶ Workstation, Server, Database, & Network Deployments
 - ▶ Windows, macOS, & Linux
- ▶ Cloud Server Management
 - ▶ Amazon Web Services & Google Cloud Platform
- ▶ Security Engineering
 - ▶ System Hardening
 - ▶ Firewall Configuration
 - ▶ Secure Web Services
 - ▶ Database Server Administration

Academic Background:

- ▶ University of Nebraska Omaha
 - ▶ Bachelor of Science
 - ▶ Computer Science
 - ▶ Mathematics
- ▶ Bellevue University
 - ▶ Master of Science (In Progress)
 - ▶ Cybersecurity
 - ▶ 27 of 36 Hours Completed

Current System Overview



Physical Security

Solid physical protection for the organization's most sensitive equipment and data storage.

▶ Current Strengths:

- ▶ Fencing
- ▶ Cameras
- ▶ Alarms
- ▶ Full-time security staff
- ▶ smart card access points
- ▶ Redundant and backup power systems
- ▶ Secure building construction

▶ Current Vulnerabilities:

- ▶ Information Disclosure
- ▶ Tampering

Wired Network Infrastructure

Three layer wired network infrastructure consisting primarily of Cisco hardware connecting ethernet and fiber-optic networks.

▶ Current Strengths:

- ▶ Reliable Cisco hardware
- ▶ Fast ethernet and fiber-optic networks
- ▶ Redundancy for layers 1 and 2

▶ Current Vulnerabilities:

- ▶ Information Disclosure
- ▶ Tampering
- ▶ Denial of Service

Wireless Connectivity

Convenient Wi-Fi access in support of a bring your own device (BYOD) policy.

▶ Current Strengths

- ▶ Wireless connectivity across campus
- ▶ Reliable Aruba Networks hardware
- ▶ Built-in firewall
- ▶ Supports a bring your own device (BYOD) policy

▶ Current Vulnerabilities:

- ▶ Information Disclosure
- ▶ Tampering
- ▶ Denial of Service

Internet

Reliable and fast broadband access provided by Comcast.

▶ Current Strengths

- ▶ Ethernet connectivity across campus buildings
- ▶ Reliable service provided by Comcast
- ▶ Redundancy
- ▶ High-speeds

▶ Current Vulnerabilities:

- ▶ Information Disclosure

Security Appliances

Firewalls provide Internet and email protection.

- ▶ Current Strengths

- ▶ Firewalls provide Internet and email protection.

- ▶ Current Vulnerabilities:

- ▶ Information Disclosure
 - ▶ Tampering
 - ▶ Denial of Service

Active Directory, Servers, & Web Hosting

Backend infrastructure meets organizational needs.

▶ Current Strengths

- ▶ Supports business activities
- ▶ Active Directory domain with group policy
- ▶ Sufficient capacity of the Storage Area Network (SAN)
- ▶ Good performance of servers and virtual machines

▶ Current Vulnerabilities:

- ▶ Information Disclosure
- ▶ Tampering
- ▶ Denial of Service

Off Campus Franchise Restaurants

Backend infrastructure meets organizational needs.

▶ Current Strengths

- ▶ Reliable Internet connections
- ▶ VPN connections to headquarters
- ▶ Antivirus and Firewall protection

▶ Current Vulnerabilities:

- ▶ Information Disclosure
- ▶ Tampering
- ▶ Denial of Service

Room for Improvement

- ▶ Some physical security and information security controls are in place
- ▶ Policies and procedures have room for improvement
- ▶ Infrastructure and security controls can be upgraded and streamlined
- ▶ Vulnerabilities identified provide a guide prioritizing improvements
- ▶ GOALS:
 - ▶ Reduce risk
 - ▶ Strengthen security posture
 - ▶ Minimize incidents
 - ▶ Maintain performance levels
 - ▶ Support business needs

Recommendations

Firewalls, Intrusion Detection Systems (IDS), and Virtual LANS (VLANS).

- ▶ Firewalls
 - ▶ Filter traffic between Campus Wireless, Campus Buildings, and Campus Core Network
 - ▶ Filter external server connection point traffic to and from the public Internet
- ▶ Intrusion Detection Systems (IDS)
 - ▶ Sophisticated security checks as information travels across networks
 - ▶ Network IDS (NIDS)
 - ▶ Wireless IDS (WIDS)
- ▶ VLANS
- ▶ Disable unused protocols and services
- ▶ Enable encryption and authentication technologies

Recommendations

Antivirus, Malware & Ransomware Protection.

- ▶ Leverage existing Dell SonicWall firewall devices
 - ▶ SSL VPN connections: *NetExtender* & *Mobile Connect* (Windows, Linux, macOS, iOS, & Android)
 - ▶ Restrict remote logons to specific IP addresses for intrusion prevention
 - ▶ Encrypted and authenticated connections protect against network intrusion
 - ▶ Supports the organization's bring your own device (BYOD) priorities
- ▶ Intrusion prevention system (IPS) enabled on NIDS appliances
 - ▶ Actively block traffic with characteristics of intrusion
 - ▶ Scanning of perimeter traffic for suspicious activities
- ▶ Host-based scanners & behavior blocking software on: servers, workstations, & mobile devices
 - ▶ Reduce vulnerability to spyware, Trojan horse, rootkit, & other malware infections

Recommendations

Email Security

- ▶ Leverage existing domain controllers
 - ▶ Domain-based authentication techniques
 - ▶ Digital signatures over email to protect against:
 - ▶ Unauthorized spam activity
 - ▶ Email domain spoofing
 - ▶ Email address forgery
 - ▶ Malware botnets
 - ▶ Phishing attempts
 - ▶ TLS encrypted email between servers
 - ▶ Prevent modification & monitoring
- ▶ Additional mail servers
 - ▶ Reduce risk of a denial of service (DOS) attack

Recommendations

Optimize network infrastructure & enhance security.

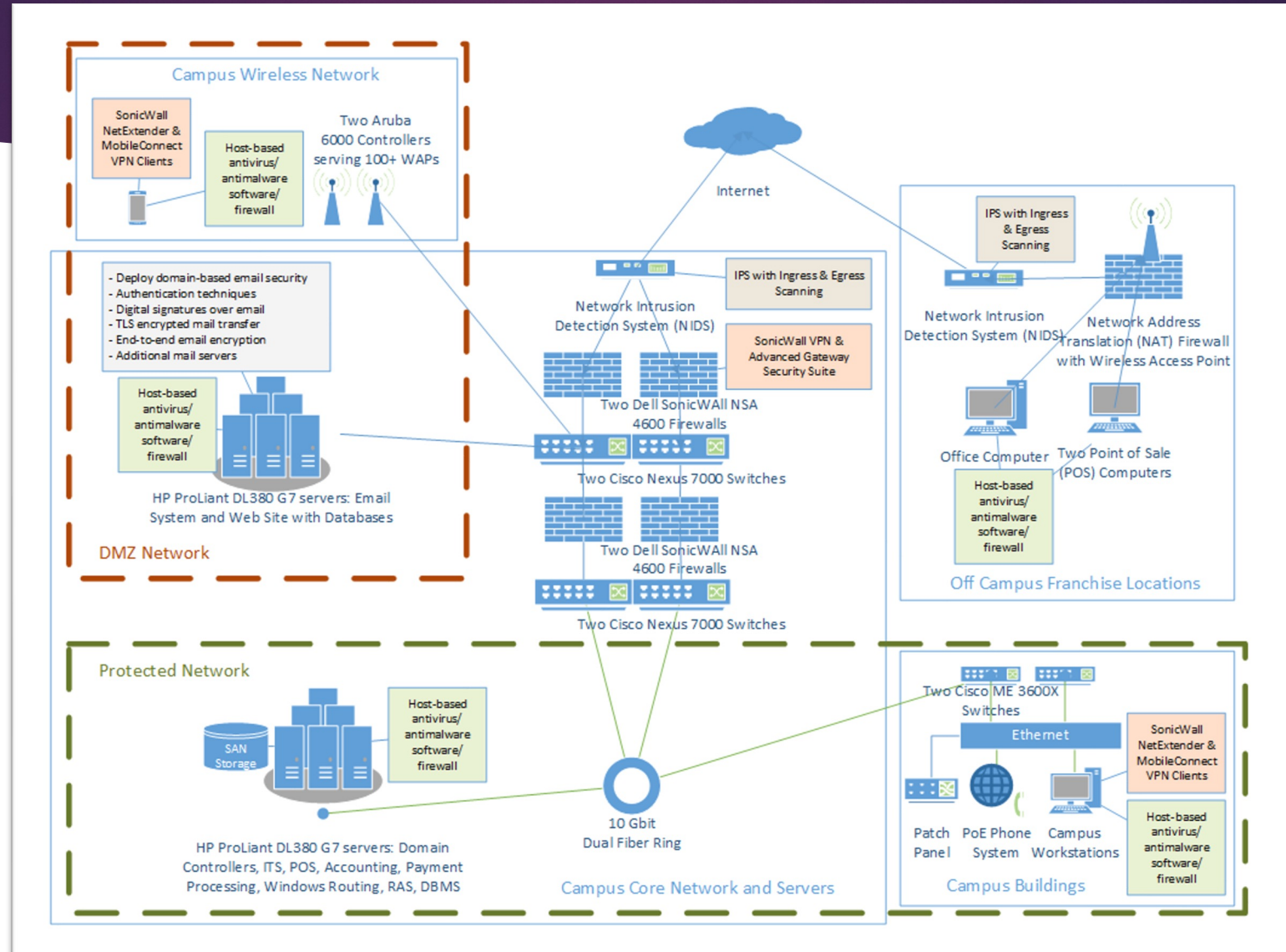
- ▶ SonicWall Advanced Gateway Security Suite (AGSS)
 - ▶ Advanced antivirus and spam filtering capabilities
 - ▶ Gateway Anti-Virus
 - ▶ Intrusion Prevention
 - ▶ Anti-Spyware
 - ▶ Content Filtering
 - ▶ Botnet Filter
 - ▶ Geo IP Filter
 - ▶ Application Firewall
 - ▶ Deep Packet Inspection of SSL (DPI-SSL)
 - ▶ Deep Packet Inspection of SSH (DPI-SSH)
 - ▶ Capture Advanced Threat Protection (ATP)

Recommendations

Optimize network infrastructure & enhance security.

- ▶ Establish a demilitarized zone (DMZ)
 - ▶ Externally accessible servers and Wireless network
 - ▶ Less trusted networks separated from critical network servers and workstations
 - ▶ Protects sensitive resources from attacks that reach the DMZ systems :
 - ▶ Rootkits
 - ▶ Bots
 - ▶ Worms
 - ▶ Malware
 - ▶ Ransomware
- ▶ Host-based firewalls
 - ▶ Enabled on servers and workstations

Implementation Proposal

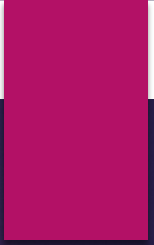


Conclusion

Aaron Martinez Solutions

Collaborative approach to implementation:

- ▶ Meet with stakeholders
 - ▶ Answer questions
 - ▶ Determine if recommendations meet Harry & Mae's expectations
 - ▶ Customize for your unique risk tolerance
 - ▶ Implementation planning
 - ▶ Upfront cost estimates
- ▶ Maintain communication throughout the process
- ▶ Support & maintenance



Q&A

Aaron Martinez Solutions

► Questions?

References

- ▶ Gu, Q., Liu, P., & Chu, C. (2004, May). Hacking Techniques in Wired Networks. Retrieved from <https://s2.ist.psu.edu/paper/hack-wired-network-may-04.pdf>
- ▶ Haron, M. (2002, October 4). Is Your Storage Area Network Secure? An Overview of Storage Area Network from Security Perspective. Retrieved from <https://www.sans.org/reading-room/whitepapers/storage/paper/516>
- ▶ Lane, H. (May 17, 2005). Security Vulnerabilities and Wireless LAN Technology. SANS. Retrieved from <https://www.sans.org/reading-room/whitepapers/wireless/paper/1629>
- ▶ Matthews, J. (2016, August 1). A Secure Approach to Deploying Wireless Networks. SANS Institute. Retrieved from <https://www.sans.org/reading-room/whitepapers/wireless/paper/37342>
- ▶ MTI. (2017, November 07). IT Security Vulnerability – Active Directory. Retrieved from <https://mti.com/blog/2017/11/07/security-vulnerabilities-active-directory/>
- ▶ SonicWall. (2019, June 07). How to Configure the SSL-VPN Feature for Use with NetExtender or Mobile Connect. Retrieved from <https://www.sonicwall.com/support/knowledge-base/how-to-configure-the-ssl-vpn-feature-for-use-with-netextender-or-mobile-connect/170505401898786/>
- ▶ SonicWall. (2019). SonicWall Capture Advanced Threat Protection (ATP). Retrieved from <https://www.sonicwall.com/products/firewalls/security-services/capture-advanced-threat-protection/>
- ▶ Stallings, W. (2017). Network Security Essentials: Applications and Standards (6th ed.). Pearson Education.
- ▶ Witcher, K. (2005, October 5). Fiber Optics and its Security Vulnerabilities. Retrieved from <https://www.sans.org/reading-room/whitepapers/physical/paper/1648>